

Lecture 14/Module 2

Transient Analysis of First and Second Order Circuit

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Today's Topics

- Singularity Function, Exercise
- Step Response – RC Circuit

Singularity Functions

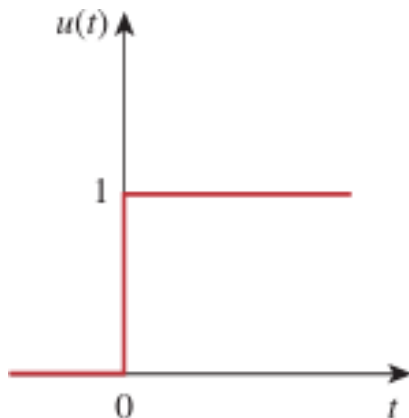
Singularity functions are also called switching functions.

Singularity functions are functions that either are discontinuous or have discontinuous derivatives.

The three most widely used singularity functions in circuit analysis are the **unit step**, the **unit impulse**, and the **unit ramp** functions.

Unit Step:

The unit step function $u(t)$ is 0 for negative values of t and 1 for positive values of t .



Mathematical equation,

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

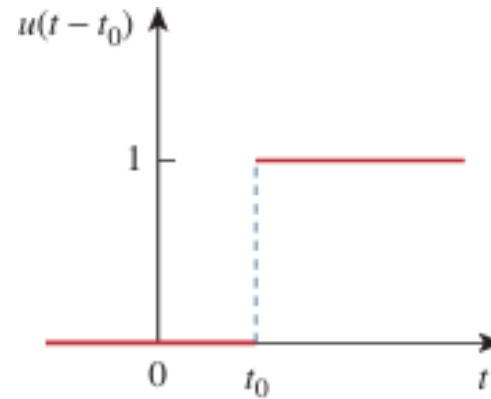
Singularity Functions

The unit step function is undefined at $t = 0$ where it changes abruptly from 0 to 1.

If the abrupt change occurs at $t = t_0$ (where t_0 is *greater than zero*) instead of $t = 0$ the unit step function becomes,

$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$$

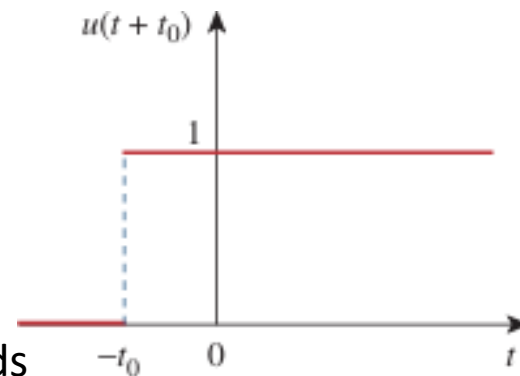
This means the same as saying that the unit step is delayed by t_0 seconds.



If the change is at $t = -t_0$, the unit step function becomes,

$$u(t + t_0) = \begin{cases} 0, & t < -t_0 \\ 1, & t > -t_0 \end{cases}$$

This means that the step has advanced by t_0 seconds



Singularity Functions

We use the step function to represent an abrupt change in voltage or current, like the changes that occur in the circuits of control systems and digital computers.

For example, if the voltage $v(t)$ is mathematically expressed as,

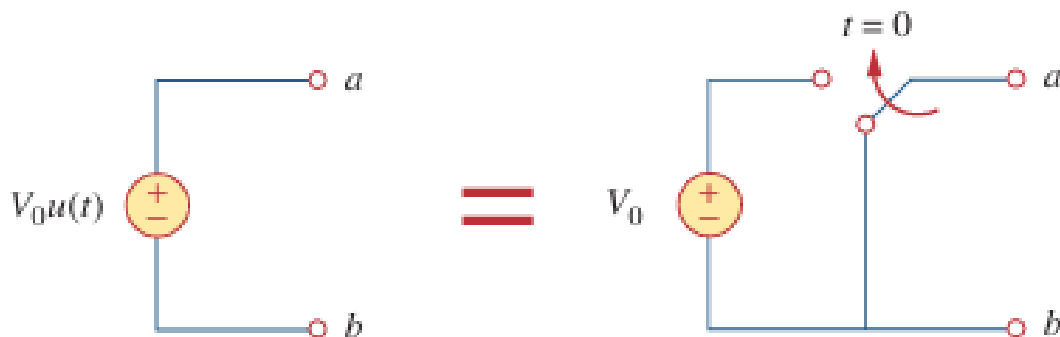
$$v(t) = \begin{cases} 0, & t < t_0 \\ V_0, & t > t_0 \end{cases}$$

This may be expressed in terms of the unit step function as,

$$v(t) = V_0 u(t - t_0)$$

If we let $t_0 = 0$, then $v(t)$ is simply the step voltage $V_0 u(t)$.

A voltage source $V_0 u(t)$ and its equivalent is shown as,

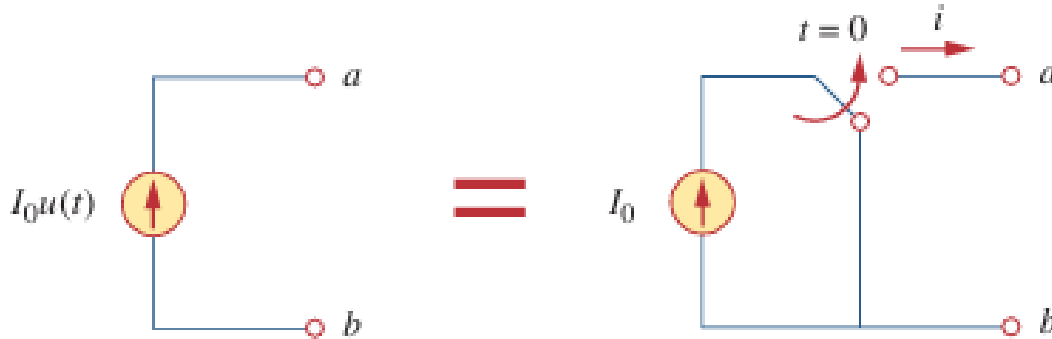


terminals a-b are short circuited for t less than zero and $v = 0$.

For t greater than 0, $v = V_0$.

Singularity Functions

Similarly, a current source of $I_0 u(t)$ is shown,



Notice that for t less than zero, there is an open circuit ($i = 0$), and that $I = I_0$ for t greater than zero.

Delta Function/Impulse function:

The derivative of the unit step function $u(t)$ is the **unit impulse function** denoted by $\delta(t)$. The unit impulse function—also known as the **delta function**.

The unit impulse function $\delta(t)$ is zero everywhere except at $t = 0$, where it is undefined.

$$\delta(t) = \frac{d}{dt}u(t) = \begin{cases} 0, & t < 0 \\ \text{Undefined}, & t = 0 \\ 0, & t > 0 \end{cases}$$

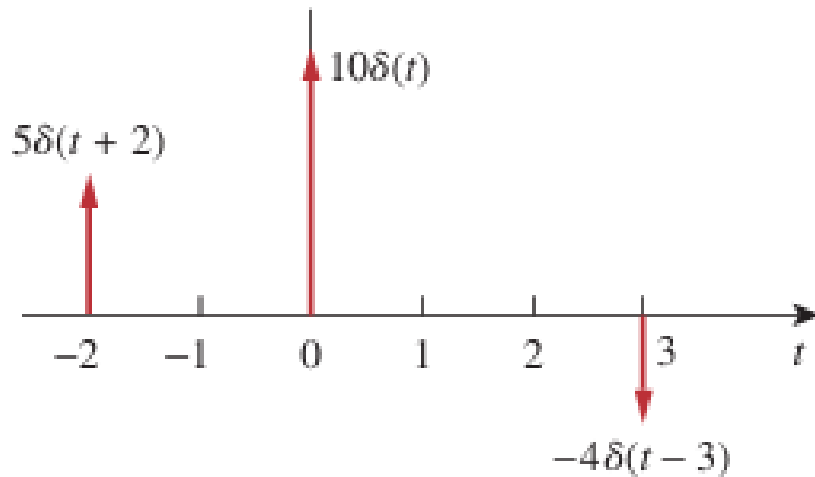
Singularity Functions

Impulsive currents and voltages occur in electric circuits as a result of switching operations or impulsive sources. Although the unit impulse function is not physically realizable (just like ideal sources, ideal resistors, etc.), it is a very useful mathematical tool.

It may be visualized as a very short duration pulse of unit area,
The unit area is known as the strength of the impulse function.

$$\int_{0^-}^{0^+} \delta(t) dt = 1$$

When an impulse function has a strength other than unity, the area of the impulse is equal to its strength. For example an impulse function has an area of 10.



Singularity Functions

To see how the impulse function affects other functions, let us evaluate the integral,

$$\int_a^b f(t)\delta(t - t_0)dt \quad \text{where } a < t_0 < b.$$

Since $\delta(t-t_0) = 0$ except at $t = t_0$ the integrand is zero except at $t = t_0$.

$$\begin{aligned}\int_a^b f(t)\delta(t - t_0)dt &= \int_a^b f(t_0)\delta(t - t_0)dt \\ &= f(t_0) \int_a^b \delta(t - t_0)dt = f(t_0)\end{aligned}$$

$$\boxed{\int_a^b f(t)\delta(t - t_0)dt = f(t_0)}$$

The special case when $t = t_0$

$$\int_{0^-}^{0^+} f(t)\delta(t)dt = f(0)$$

This shows that when a function is integrated with the impulse function, we obtain the value of the function at the point where the impulse occurs. This is a highly useful property of the impulse function known as the sampling property.

Singularity Functions

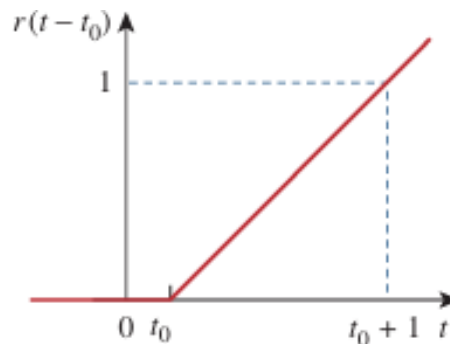
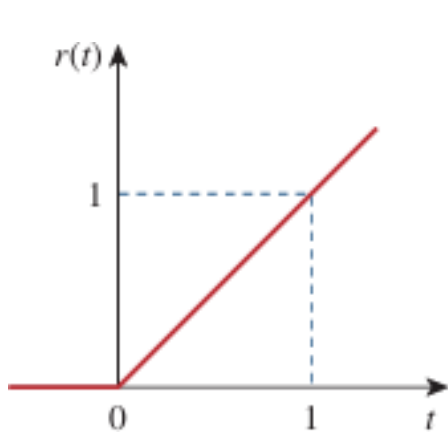
Unit Ramp Function

Integrating the unit step function results in the unit ramp function $r(t)$.

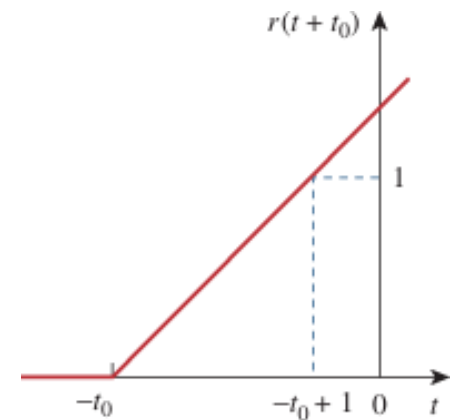
$$r(t) = \int_{-\infty}^t u(\lambda) d\lambda = tu(t) \quad \text{or}$$

$$r(t) = \begin{cases} 0, & t \leq 0 \\ t, & t \geq 0 \end{cases}$$

The unit ramp function is zero for negative values of t and has a unit slope for positive values of t . The unit ramp function may be delayed or advanced.



$$r(t - t_0) = \begin{cases} 0, & t \leq t_0 \\ t - t_0, & t \geq t_0 \end{cases}$$



$$r(t + t_0) = \begin{cases} 0, & t \leq -t_0 \\ t + t_0, & t \geq -t_0 \end{cases}$$

Singularity Functions

The three singularity functions (impulse, step, and ramp) are related by differentiation as

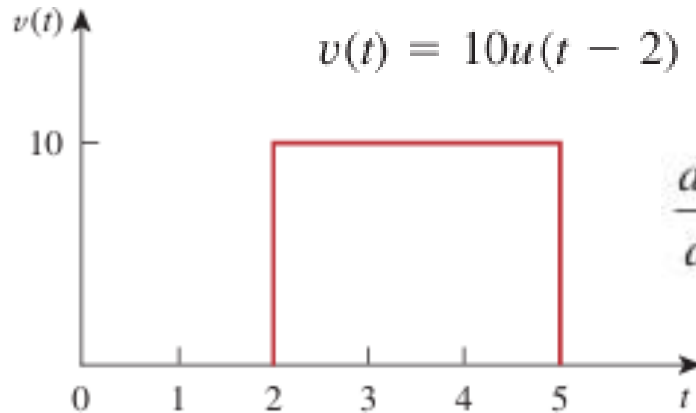
$$\delta(t) = \frac{du(t)}{dt}, \quad u(t) = \frac{dr(t)}{dt}$$

The three singularity functions (impulse, step, and ramp) are related by integration as

$$u(t) = \int_{-\infty}^t \delta(\lambda) d\lambda, \quad r(t) = \int_{-\infty}^t u(\lambda) d\lambda$$

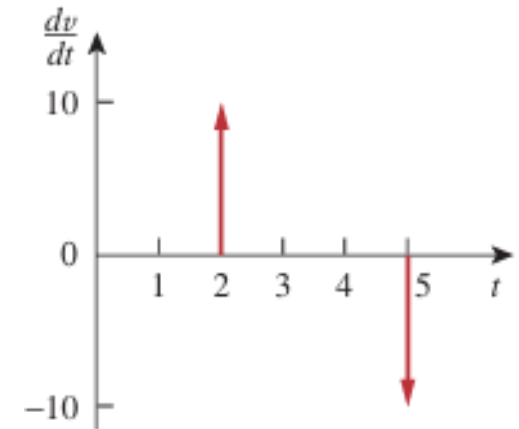
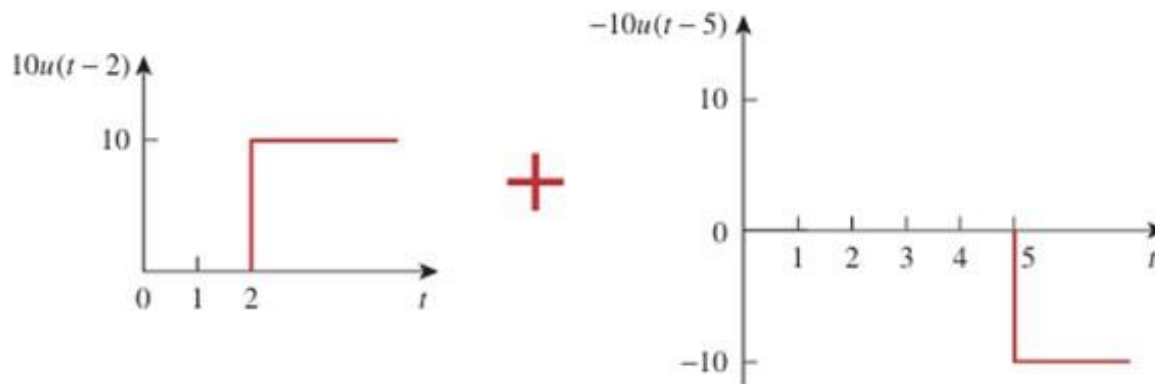
Exercise

1. Express the voltage pulse in Figure in terms of the unit step. Calculate its derivative and sketch it.



$$v(t) = 10u(t - 2) - 10u(t - 5) = 10[u(t - 2) - u(t - 5)]$$

$$\frac{dv}{dt} = 10[\delta(t - 2) - \delta(t - 5)]$$

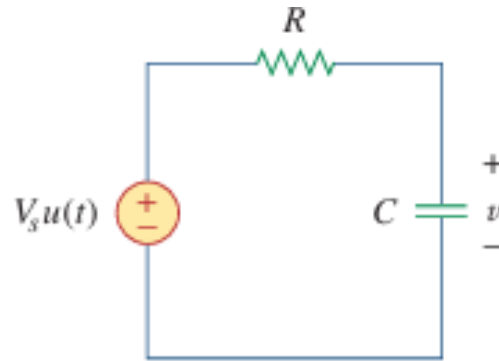
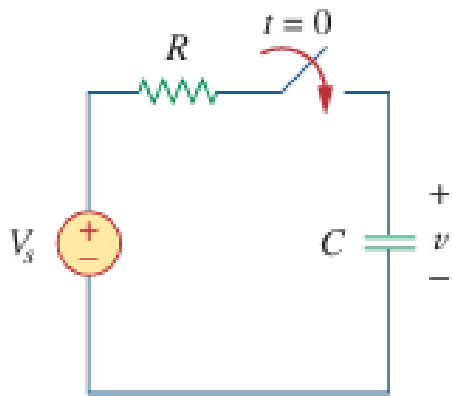


Derivative is impulse

Step Response of RC circuit

The step response of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.

When the dc source of an RC circuit is suddenly applied, the voltage or current source can be modeled as a step function.



Initial voltage on the capacitor is V_0

$$v(0^-) = v(0^+) = V_0$$

Where $v(0^-)$ is the voltage across the capacitor just before switching and $v(0^+)$ is its voltage immediately after switching.

Applying KCL;

$$C \frac{dv}{dt} + \frac{v - V_s u(t)}{R} = 0$$

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC} u(t)$$

Where v is the voltage across the capacitor. For $t > 0$,

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC}$$

Step Response of RC circuit

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC}$$

Rearranging terms gives,

$$\frac{dv}{dt} = -\frac{v - V_s}{RC}$$

$$\frac{dv}{v - V_s} = -\frac{dt}{RC}$$

Integrating both sides and introducing the initial conditions,

$$\ln(v - V_s) \Big|_{V_0}^{v(t)} = -\frac{t}{RC} \Big|_0^t$$

$$\ln(v(t) - V_s) - \ln(V_0 - V_s) = -\frac{t}{RC} + 0$$

$$\ln \frac{v - V_s}{V_0 - V_s} = -\frac{t}{RC}$$

Step Response of RC circuit

Taking the exponential of both sides,

$$\frac{v - V_s}{V_0 - V_s} = e^{-t/\tau}, \quad \tau = RC$$
$$v - V_s = (V_0 - V_s)e^{-t/\tau}$$

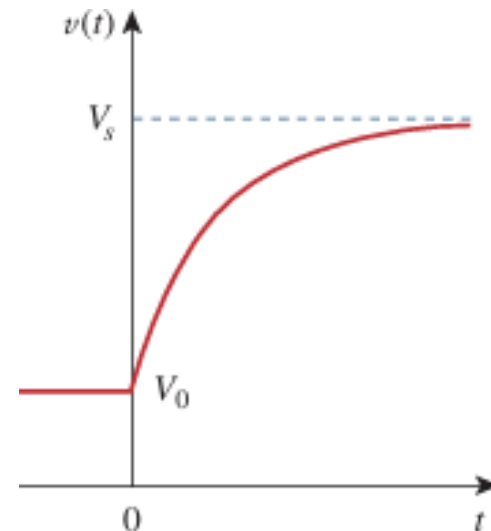
$$v(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \quad t > 0$$

Thus,

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau}, & t > 0 \end{cases}$$

Assuming that $V_s > V_0$ a plot of $v(t)$ is shown:

This is known as the **complete response** (or total response) of the RC circuit to a sudden application of a dc voltage source, **assuming the capacitor is initially charged**.



Thank You